

University of Frontier Technology, Bangladesh

Department of Cyber Security Engineering (CySE)

Lab Manual

Course title : Threat Modelling and Security Monitoring Sessional

Course code : CSE 202

Name : Windows Log Analysis with PowerShell

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WEEK 4 LAB MANUAL

Windows Log Analysis with PowerShell

1. Introduction

Windows Event Logs are important sources of information for security monitoring and investigation. They record different activities performed on a Windows computer, including successful logins, failed login attempts, account changes, and access to shared resources.

In this lab, we use Windows PowerShell to analyze security event logs and identify suspicious activities.

PowerShell provides several commands that help security analysts collect, filter, analyze, and export Windows event log information.

2. Objectives

The main objectives of this lab are:

1. To understand Windows Security Event Logs.
2. To identify successful and failed login attempts.
3. To investigate suspicious login activities.
4. To identify possible brute-force attacks.
5. To analyze logon types and authentication events.
6. To detect privilege escalation activities.
7. To monitor access to shared network resources.
8. To export security log reports into CSV and HTML formats.

3. Requirements

The following tools and requirements are needed:

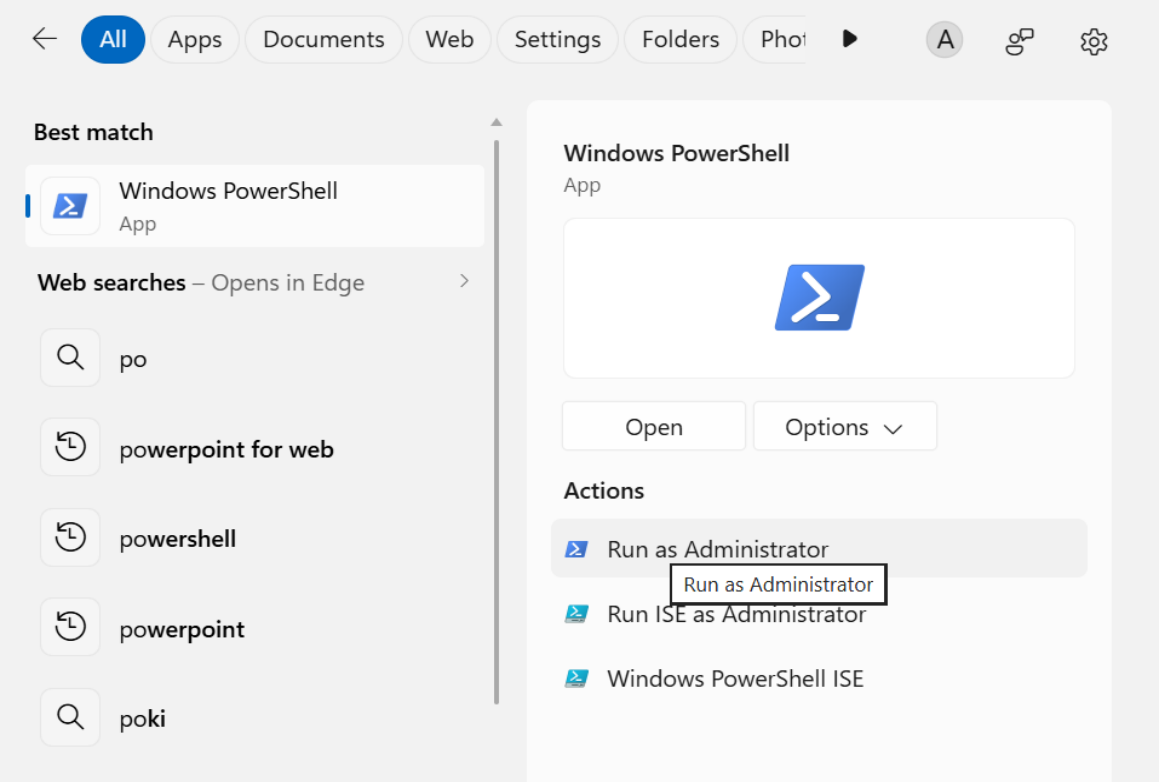
- Windows 10 or Windows 11.
- Windows PowerShell.
- Administrator privileges.
- Windows Security Event Logs.
- Basic knowledge of PowerShell commands.

4. Important Windows Security Event IDs

Event ID	Description
4624	Successful Logon
4625	Failed Logon
4648	Logon Attempt Using Explicit Credentials
4672	Special Privileges Assigned to a New Logon
4732	Member Added to a Security-Enabled Local Group
5140	Network Share Accessed

5. Lab Exercises and Commands

Open PowerShell as Administrator: To open PowerShell with administrator privileges.

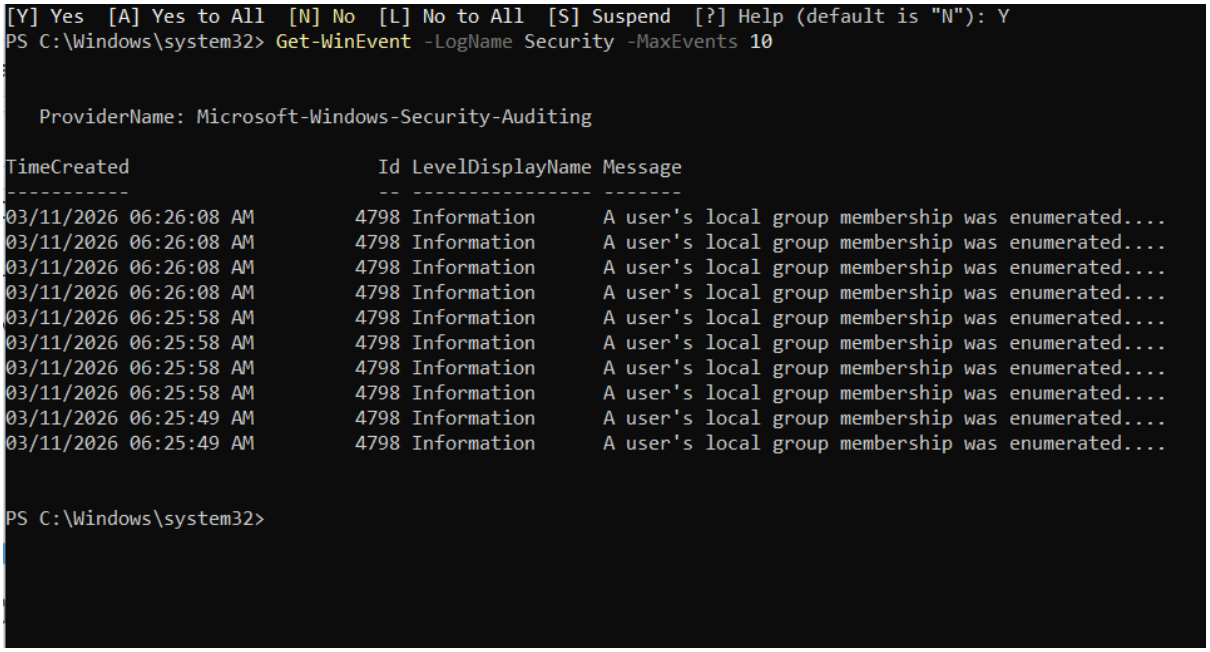


Exercise 1: View Recent Security Events

Objective: To view the 10 most recent Windows Security Events.

Command:

Get-WinEvent -LogName Security -MaxEvents 10



Explanation:

- **Get-WinEvent** retrieves Windows event logs.
- **-LogName** Security specifies the Security log.
- **-MaxEvents** 10 displays the 10 most recent events.

Exercise 2: Filter by Event ID

Objective: To view the last 20 successful logon events (Event ID 4624).

Command:

Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4624} -MaxEvents 20

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4624} -MaxEvents 20


   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
03/11/2026 06:25:45 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:20:45 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:20:07 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:20:01 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:19:52 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:19:03 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:17:15 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:15:54 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:15:45 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:43 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:43 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:42 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:41 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:41 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:41 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:41 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:41 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:41 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:41 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:13:41 AM        4624 Information      An account was successfully logged on....
```

Explanation:

- **-FilterHashtable** filters events using specific conditions.
- **LogName='Security'** selects the Security log.
- **ID=4624** filters successful logon events.
- **-MaxEvents 20** displays the latest 20 events.

Exercise 3: Filter by Time Range

Objective: To retrieve security events from the last 24 hours.

Command:

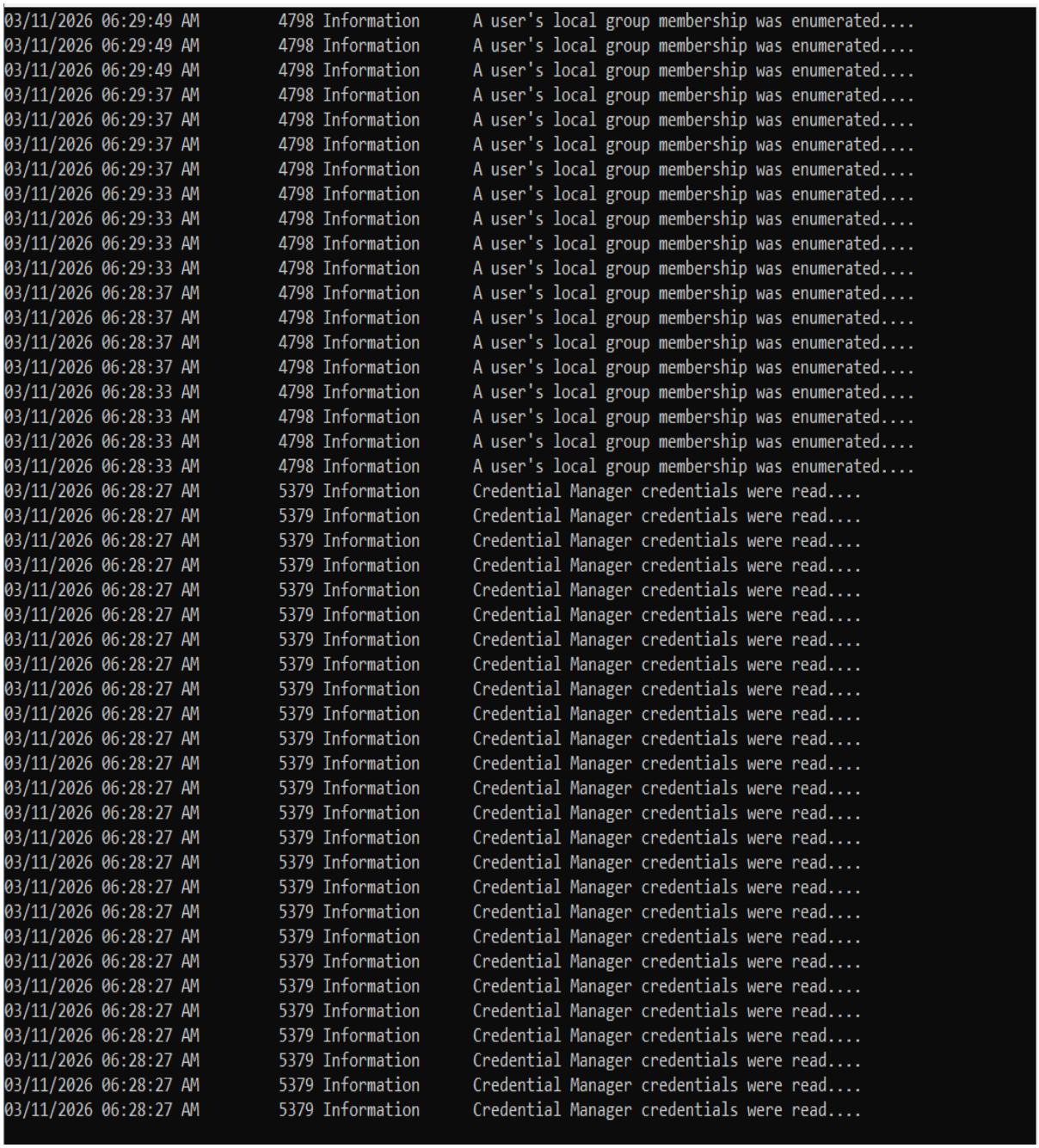
$StartTime = (Get-Date).AddDays(-1)

Get-WinEvent -FilterHashtable @{LogName='Security'; StartTime=$StartTime} -MaxEvents 100

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; StartTime=$StartTime} -MaxEvents 100


   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
03/11/2026 06:32:54 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:54 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:54 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:54 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:33 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:33 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:33 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:33 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:08 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:08 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:08 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:32:08 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:58 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:58 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:58 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:58 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:49 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:49 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:49 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:49 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:33 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:33 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:33 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:31:33 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:30:45 AM        4672 Information      Special privileges assigned to new logon....
03/11/2026 06:30:45 AM        4624 Information      An account was successfully logged on....
03/11/2026 06:30:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:30:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:30:37 AM        4798 Information      A user's local group membership was enumerated....
03/11/2026 06:30:37 AM        4798 Information      A user's local group membership was enumerated....
```



Explanation:

- **Get-Date** gets the current date and time.
- **AddDays(-1)** subtracts one day.
- **StartTime** specifies the starting time.
- **-MaxEvents 100** displays up to 100 events.

Exercise 4: Count Events

Objective: To count the failed logon attempts (Event ID 4625) that occurred in the last 24 hours.

Command:

Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=$StartTime}

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=$StartTime}


   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
03/11/2026 06:08:04 AM        4625 Information      An account failed to log on....
03/11/2026 06:05:50 AM        4625 Information      An account failed to log on....
02/11/2026 11:41:06 PM        4625 Information      An account failed to log on....
02/11/2026 11:39:21 PM        4625 Information      An account failed to log on....
02/11/2026 11:39:09 PM        4625 Information      An account failed to log on....
02/11/2026 09:57:45 PM        4625 Information      An account failed to log on....
02/11/2026 09:57:31 PM        4625 Information      An account failed to log on....
02/11/2026 09:14:45 PM        4625 Information      An account failed to log on....
02/11/2026 09:14:41 PM        4625 Information      An account failed to log on....
02/11/2026 09:14:20 PM        4625 Information      An account failed to log on....
02/11/2026 09:08:03 PM        4625 Information      An account failed to log on....
02/11/2026 09:04:27 PM        4625 Information      An account failed to log on....
```

(Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=$StartTime} -ErrorAction SilentlyContinue).Count

```
PS C:\Windows\system32> (Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=$StartTime} -ErrorAc
tion SilentlyContinue).Count
12
```

Explanation:

- **ID=4625** selects failed logon events.

Exercise 5: Understanding Event Structure

Objective: To examine the structure and properties of a Windows Security Event.

Command:

$Event = Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625} -MaxEvents 1

$Event | Format-List

```
PS C:\Windows\system32> $Event = Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625} -MaxEvents 1
PS C:\Windows\system32> $Event | Format-List *


Message              : An account failed to log on.

                       Subject:
                        Security ID:            S-1-5-21-1695193204-1341121097-2690481908-1001
                        Account Name:           Anika Akter Nira
                        Account Domain:         DESKTOP-BUG72BE
                        Logon ID:               0x8BA8B

                       Logon Type:                      2

                       Account For Which Logon Failed:
                        Security ID:            S-1-0-0
                        Account Name:           Anika Akter Nira
                        Account Domain:         DESKTOP-BUG72BE

                       Failure Information:
                        Failure Reason:         Unknown user name or bad password.
                        Status:                 0xC000006E
                        Sub Status:             0xC000006E

                       Process Information:
                        Caller Process ID:      0x446c
                        Caller Process Name:    C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe

                       Network Information:
                        Workstation Name:       DESKTOP-BUG72BE
                        Source Network Address: -
                        Source Port:            -

                       Detailed Authentication Information:
                        Logon Process:          Advapi
                        Authentication Package: Negotiate
                        Transited Services:     -
                        Package Name (NTLM only):       -
                        Key Length:             0

                       This event is generated when a logon request fails. It is generated on the computer where access was
                       attempted.

                       The Subject fields indicate the account on the local system which requested the logon. This is most
                       commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

                       Detailed Authentication Information:
                        Logon Process:          Advapi
                        Authentication Package: Negotiate
                        Transited Services:     -
                        Package Name (NTLM only):       -
                        Key Length:             0

                       This event is generated when a logon request fails. It is generated on the computer where access was
                       attempted.

                       The Subject fields indicate the account on the local system which requested the logon. This is most
                       commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

                       The Logon Type field indicates the kind of logon that was requested. The most common types are 2
                       (interactive) and 3 (network).

                       The Process Information fields indicate which account and process on the system requested the logon.

                       The Network Information fields indicate where a remote logon request originated. Workstation name is
                       not always available and may be left blank in some cases.

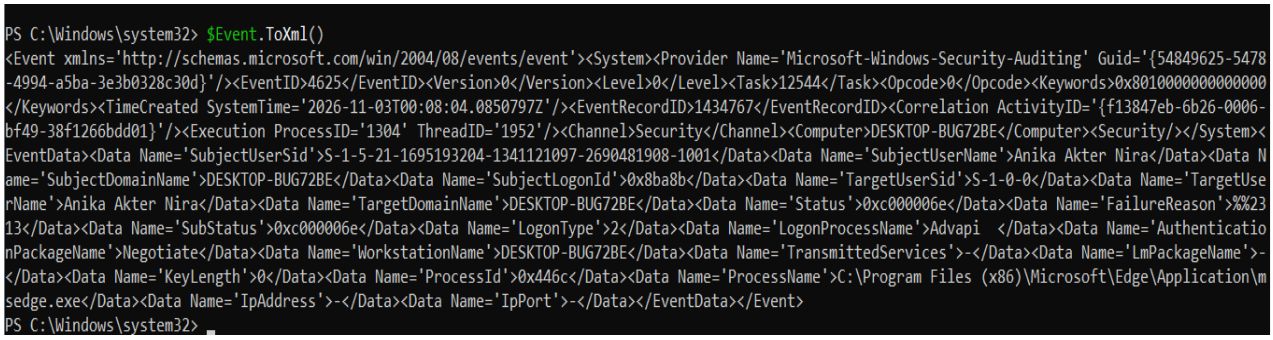
                       The authentication information fields provide detailed information about this specific logon request.
                        - Transited services indicate which intermediate services have participated in this logon request.
                        - Package name indicates which sub-protocol was used among the NTLM protocols.
                        - Key length indicates the length of the generated session key. This will be 0 if no session key was
                       requested.
Id                   : 4625
Version              : 0
Qualifiers           :
Level                : 0
Task                 : 12544
Opcode               : 0
Keywords             : -9218868437227405312
RecordId             : 1434767
ProviderName         : Microsoft-Windows-Security-Auditing
ProviderId           : 54849625-5478-4994-a5ba-3e3b0328c30d
LogName              : Security
ProcessId            : 1304
ThreadId             : 1952
MachineName          : DESKTOP-BUG72BE
UserId               :
TimeCreated          : 03/11/2026 06:08:04 AM
ActivityId           : f13847eb-6b26-0006-bf49-38f1266bdd01
RelatedActivityId    :
ContainerLog         : Security
MatchedQueryIds      : {0}
Bookmark             : System.Diagnostics.Eventing.Reader.EventBookmark
LevelDisplayName     : Information
```

Explanation:

- **$Event** stores one failed logon event.
- **Format-List** displays all available event properties.
- **TimeCreated** shows when the event occurred.
- **Id** shows the Event ID.
- **Message** displays event details.
- **Properties** contains event-specific data.

View XML Command:

$Event.ToXml()



Exercise 6: Extract Data from Failed Logons

Objective: To extract important information from failed logon events.

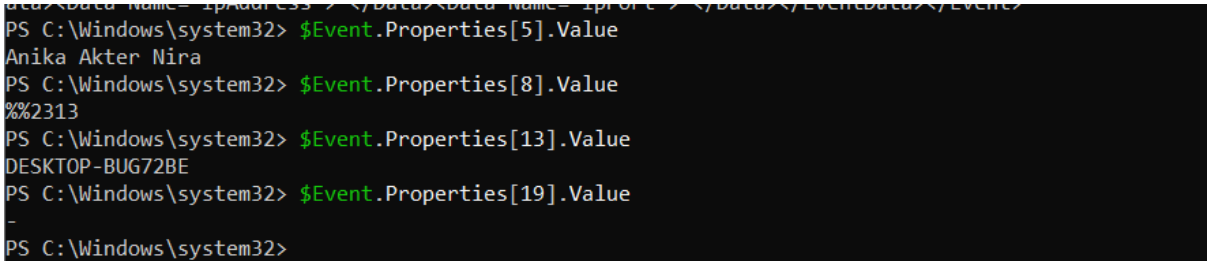
Command:

$Event.Properties[5].Value

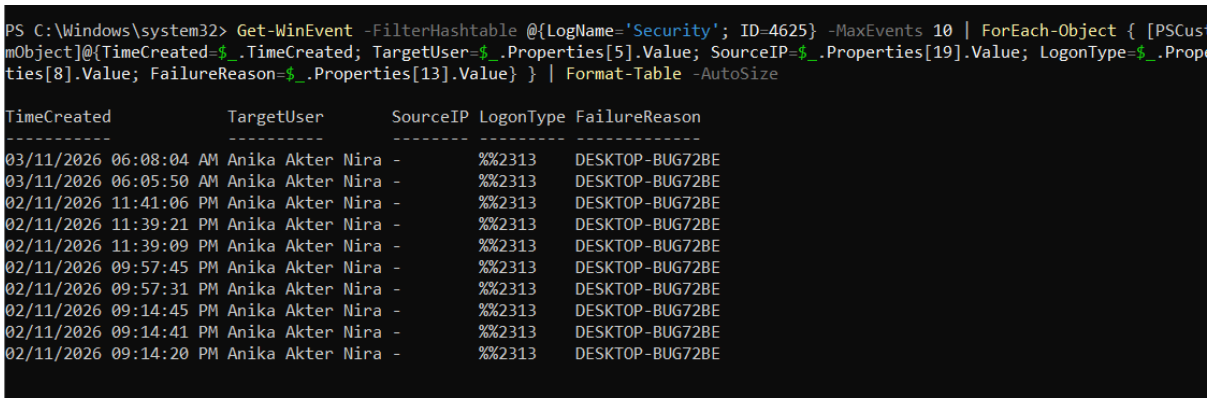
$Event.Properties[8].Value

$Event.Properties[13].Value

$Event.Properties[19].Value



Get-WinEvent -FilterHashtable @{ LogName='Security' ID=4625} -MaxEvents 10 |ForEach-Object { [PSCustomObject]@{ TimeCreated = $_.TimeCreated TargetUser = $_.Properties[5].Value SourceIP = $_.Properties[19].Value LogonType = $_.Properties[8].Value FailureReason = $_.Properties[13].Value }} | Format-Table –AutoSize



Explanation:

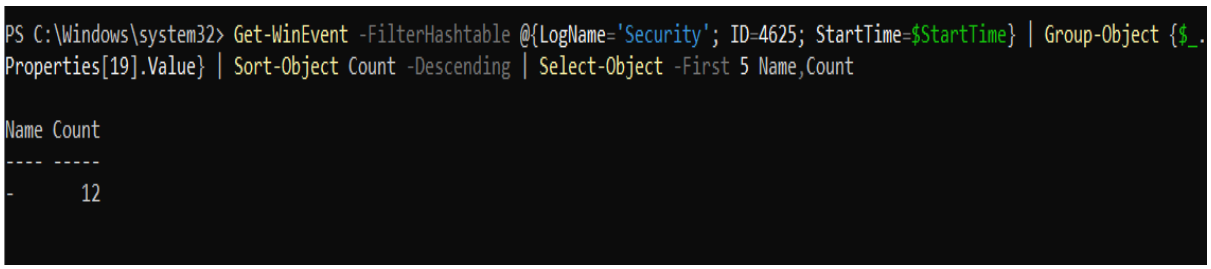
- **ID=4625** selects failed logon events.

Exercise 7: Analyze Failed Logon Attempts

Objective: To identify the top 5 source IP addresses responsible for failed logon attempts.

Command:

Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=$StartTime} | Group-Object {$_.Properties[19].Value} | Sort-Object Count -Descending | Select-Object -First 5 Name,Count



Explanation:

- Retrieves failed logon events from the last 24 hours.

- Extracts source IP addresses.
- Removes empty IP address values.
- **Group-Object** groups identical IP addresses.
- **Sort-Object** sorts them by frequency.
- **Select-Object -**First 5 displays the top 5 IP addresses.

Exercise 8: Identify Targeted Accounts

Objective: To identify the usernames targeted in failed logon attempts.

Command:

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=$StartTime} | Group-Object {$_.
Properties[8].Value} | Sort-Object Count -Descending | Select-Object Name,Count

Name   Count
----   -----
%%2313    12
```

Explanation:

- Retrieves failed logon events.
- Extracts targeted usernames.
- Groups identical usernames.
- Counts failed login attempts for each username.

Exercise 9: Analyze Logon Types

Objective: To analyze the distribution of different logon types in failed login attempts.

Command:

Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=(Get-Date).AddDays(-1)} | ForEach-Object { $LogonType=$_.Properties[8].Value; switch ($LogonType) { 2 {'Interactive'} 3 {'Network'} 4 {'Batch'} 5 {'Service'} 7 {'Unlock'} 8 {'NetworkCleartext'} 9 {'NewCredentials'} 10 {'RemoteInteractive'} 11 {'CachedInteractive'} default {"Unknown ($LogonType)"} } } | Group-Object | Sort-Object Count -

Descending | Select-Object @{Name='Logon Type';Expression={$_.Name}}, @{Name='Count';Expression={$_.Count}} | Format-Table -AutoSize

```
PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{
>> LogName = 'Security'
>> ID = 4625
>> StartTime = (Get-Date).AddDays(-1)
>> } | ForEach-Object {
>> $_.Properties[5].Value
>> } | Where-Object { $_ -ne '-' -and $_ -ne '' } |
>> Group-Object |
>> Sort-Object Count -Descending |
>> Select-Object -First 5 `
>> @{Name='Username';Expression={$_.Name}},
>> @{Name='Failed Attempts';Expression={$_.Count}} |
>> Format-Table -AutoSize
>>
>>

Username  Failed Attempts
--------  ---------------
Microsoft               3


PS C:\WINDOWS\system32> Get-WinEvent -FilterHashtable @{
>>
>>     LogName   = 'Security'
>>
>>     ID        = 4625
>>
>>     StartTime = (Get-Date).AddDays(-1)
>>
>> } | ForEach-Object {
>>
>>     $LogonType = [int]$_.Properties[8].Value
>>
>>     switch ($LogonType) {
>>
>>         2  { 'Interactive' }
>>
>>         3  { 'Network' }
>>
>>         4  { 'Batch' }
>>
>>         5  { 'Service' }
>>
>>         7  { 'Unlock' }
>>
>>         8  { 'NetworkCleartext' }
>>
>>         9  { 'NewCredentials' }
>>
>>         10 { 'RemoteInteractive' }
>>
>>         11 { 'CachedInteractive' }
>>
>>         default { "Unknown ($LogonType)" }
>>
>>     }
>>
>> } | Group-Object |
>>
>>     Sort-Object Count -Descending |
>>
```

```
Count Name                      Group
----- ----                      -----
    3 Unknown ()                {Unknown (), Unknown (), Unknown ()}



Name       Value
----       -----
Expression $_.Name
Name       Logon Type
Expression $_.Count
Name       Count
```

Explanation:

- Retrieves failed logon events from the last 24 hours.
- Extracts the logon type.

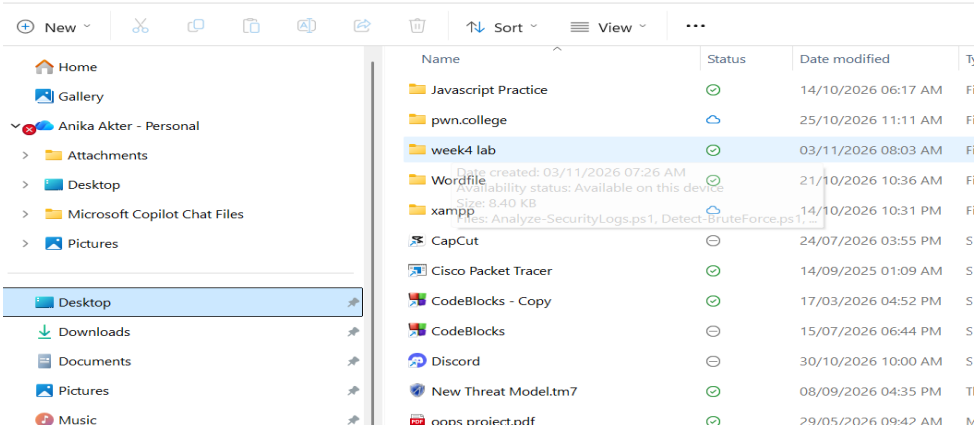
- switch identifies the name of each logon type.
- Group-Object groups similar logon types.
- Sort-Object sorts the results by count.
- Format-Table displays the distribution.

Exercise 10: Identify Brute-Force Patterns

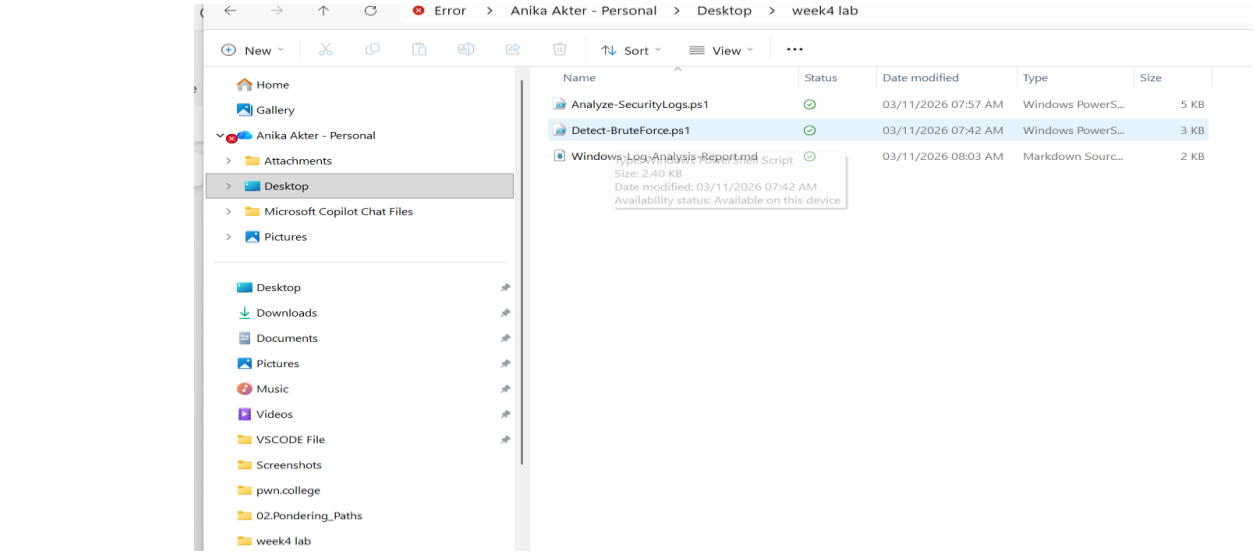
Objective: To detect possible brute-force attacks by identifying repeated failed login attempts from the same IP address.

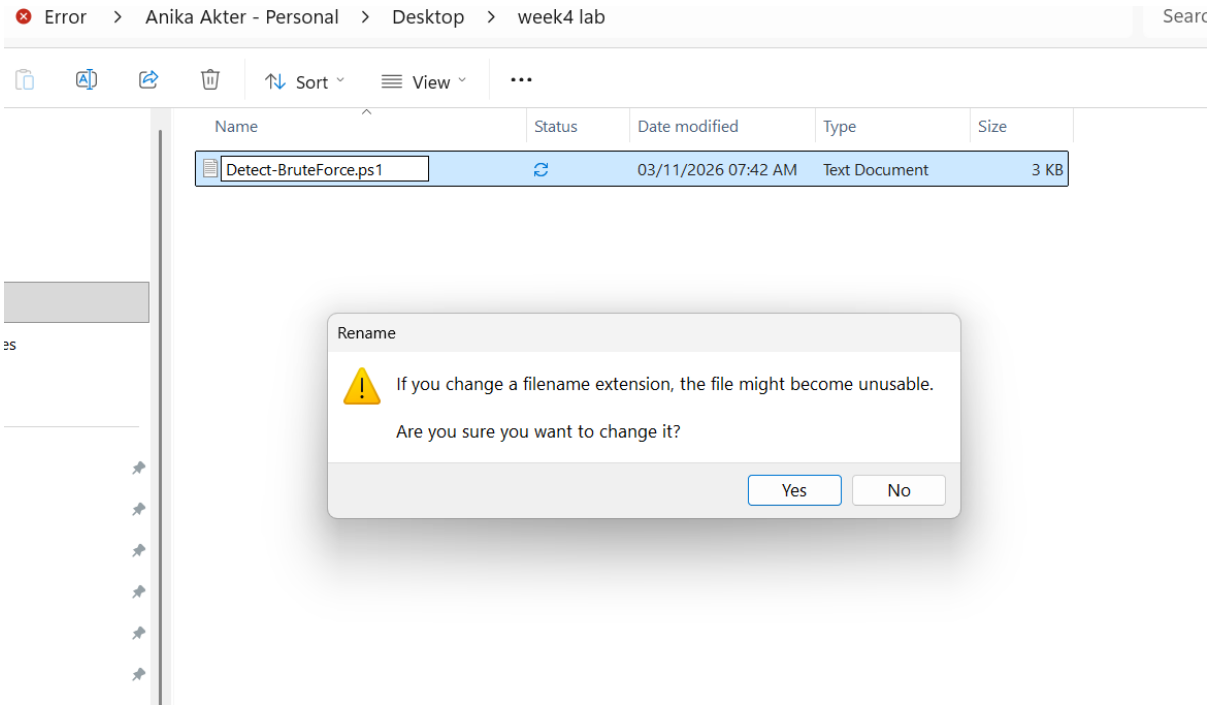
Creating Script File

1. **Create a Folder on the Desktop:**
   a. Go to your Desktop and create a new folder named:Week-4-Lab



2. **Create a File inside the Folder:**
   a. Open the Week-4-Lab folder.
   b. Create a new file named:Detect-BruteForce.ps1





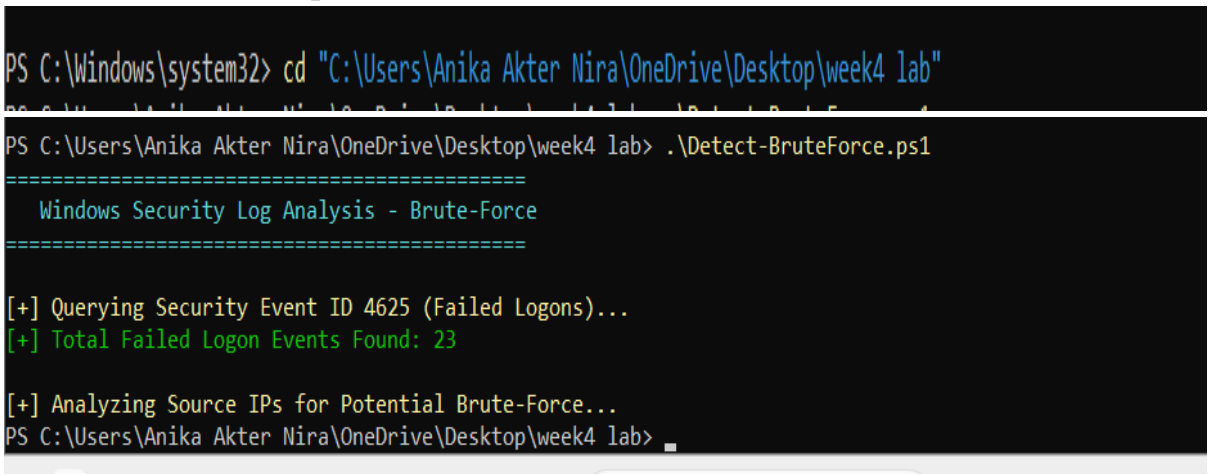
(Note: Make sure to enable **File name extensions** in Windows File Explorer so it doesn't accidentally save as Detect-BruteForce.ps1.txt).

3. **Open the File:**
    a. Open Detect-BruteForce.ps1 using **Notepad**.

Save the Script As: Detect-BruteForce.ps1

Run Command:

.\Detect-BruteForce.ps1



Exercise 12: Detect Explicit Credential Usage

Objective: To identify activities where explicit credentials are used for authentication.

Command:

Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4648; StartTime=(Get-Date).AddDays(-1)} | ForEach-Object {$xml=[xml]$_.ToXml();[PSCustomObject]@{TimeCreated=$_.TimeCreated; SourceUser=$xml.Event.EventData.Data[1].'#text'; TargetUser=$xml.Event.EventData.Data[5].'#text'; TargetServer=$xml.Event.EventData.Data[8].'#text'; Process=$xml.Event.EventData.Data[11].'#text'}} | Format-Table -AutoSize

```
PS C:\> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=5140; StartTime=(Get-Date).AddDays(-7)} |
 ForEach-Object {$xml=[xml]$_.ToXml(); [PSCustomObject]@{TimeCreated=$_.TimeCreated; Account=$xml.Event.
EventData.Data[1].'#text'; SourceIP=$xml.Event.EventData.Data[3].'#text'; ShareName=$xml.Event.EventData
.Data[4].'#text'}} | Format-Table -AutoSize
Get-WinEvent : No events were found that match the specified selection criteria.
At line:1 char:1
+ Get-WinEvent -FilterHashtable @{LogName='Security'; ID=5140; StartTim ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (:) [Get-WinEvent], Exception
    + FullyQualifiedErrorId : NoMatchingEventsFound,Microsoft.PowerShell.Commands.GetWinEventCommand

PS C:\> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=(Get-Date).AddDays(-7)} |
 ForEach-Object {$xml=[xml]$_.ToXml(); [PSCustomObject]@{TimeCreated=$_.TimeCreated; Account=($xml.Event
.EventData.Data | Where-Object {$_.Name -eq 'TargetUserName'}).'#text'; IPAddress=($xml.Event.EventData.
Data | Where-Object {$_.Name -eq 'IpAddress'}).'#text'}} | Format-Table -AutoSize

TimeCreated             Account          IPAddress
-----------             -------          ---------
03/11/2026 07:30:52 AM Anika Akter Nira -
03/11/2026 07:30:16 AM Anika Akter Nira -
03/11/2026 07:30:10 AM Anika Akter Nira -
03/11/2026 07:18:39 AM Anika Akter Nira -
03/11/2026 07:17:33 AM Anika Akter Nira -
03/11/2026 07:16:44 AM Anika Akter Nira -
03/11/2026 07:16:32 AM Anika Akter Nira -
03/11/2026 07:16:30 AM Anika Akter Nira -
03/11/2026 07:14:03 AM Anika Akter Nira -
03/11/2026 07:14:01 AM Anika Akter Nira -
03/11/2026 07:13:53 AM Anika Akter Nira -
03/11/2026 06:08:04 AM Anika Akter Nira -
03/11/2026 06:05:50 AM Anika Akter Nira -
02/11/2026 11:41:06 PM Anika Akter Nira -
02/11/2026 11:39:21 PM Anika Akter Nira -
02/11/2026 11:39:09 PM Anika Akter Nira -
02/11/2026 09:57:45 PM Anika Akter Nira -
02/11/2026 09:57:31 PM Anika Akter Nira -
02/11/2026 09:14:45 PM Anika Akter Nira -
02/11/2026 09:14:41 PM Anika Akter Nira -
02/11/2026 09:14:20 PM Anika Akter Nira -
02/11/2026 09:08:03 PM Anika Akter Nira -
02/11/2026 09:04:27 PM Anika Akter Nira -
01/11/2026 11:19:30 PM Anika Akter Nira -
```

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03/11/2026 06:08:04 AM Anika Akter Nira -
03/11/2026 06:05:50 AM Anika Akter Nira -
02/11/2026 11:41:06 PM Anika Akter Nira -
02/11/2026 11:39:21 PM Anika Akter Nira -
02/11/2026 11:39:09 PM Anika Akter Nira -
02/11/2026 09:57:45 PM Anika Akter Nira -
02/11/2026 09:57:31 PM Anika Akter Nira -
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02/11/2026 09:14:41 PM Anika Akter Nira -
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02/11/2026 09:08:03 PM Anika Akter Nira -
02/11/2026 09:04:27 PM Anika Akter Nira -
01/11/2026 11:19:30 PM Anika Akter Nira -
01/11/2026 11:17:23 PM Anika Akter Nira -
01/11/2026 09:28:50 PM Anika Akter Nira -
01/11/2026 09:28:40 PM Anika Akter Nira -
01/11/2026 09:27:58 PM Anika Akter Nira -
01/11/2026 09:27:45 PM Anika Akter Nira -
01/11/2026 01:30:25 PM Anika Akter Nira -
01/11/2026 01:10:04 PM Anika Akter Nira -
01/11/2026 01:08:07 PM Anika Akter Nira -
01/11/2026 01:07:24 PM Anika Akter Nira -
01/11/2026 01:05:33 PM Anika Akter Nira -
01/11/2026 01:04:41 PM Anika Akter Nira -
01/11/2026 01:02:43 PM Anika Akter Nira -
01/11/2026 01:01:35 PM Anika Akter Nira -
01/11/2026 01:00:52 PM Anika Akter Nira -
01/11/2026 01:00:12 PM Anika Akter Nira -
01/11/2026 12:59:38 PM Anika Akter Nira -
01/11/2026 12:58:41 PM Anika Akter Nira -
01/11/2026 12:56:53 PM Anika Akter Nira -
01/11/2026 12:54:50 PM Anika Akter Nira -
01/11/2026 12:54:09 PM Anika Akter Nira -
01/11/2026 12:53:31 PM Anika Akter Nira -
01/11/2026 12:52:13 PM Anika Akter Nira -
01/11/2026 12:51:18 PM Anika Akter Nira -
01/11/2026 12:50:17 PM Anika Akter Nira -
01/11/2026 12:48:38 PM Anika Akter Nira -
01/11/2026 12:47:53 PM Anika Akter Nira -
01/11/2026 12:47:14 PM Anika Akter Nira -
01/11/2026 12:46:38 PM Anika Akter Nira -
01/11/2026 12:46:12 PM Anika Akter Nira -
01/11/2026 12:45:36 PM Anika Akter Nira -
01/11/2026 12:44:49 PM Anika Akter Nira -
01/11/2026 12:44:10 PM Anika Akter Nira -
01/11/2026 12:43:48 PM Anika Akter Nira -
01/11/2026 12:41:04 PM Anika Akter Nira -
01/11/2026 12:40:54 PM Anika Akter Nira -
01/11/2026 12:39:48 PM Anika Akter Nira -
```

Explanation:

- ID=4648 identifies explicit credential usage events.
- ToXml() converts the event into XML format.
- Extracts source user and target user information.
- Identifies the target server and process.
- Displays the extracted information in a table.

Exercise 13: Detect Network Share Access

Objective: To monitor network share access and investigate possible lateral movement.

```
    + FullyQualifiedErrorId : NoMatchingEventsFound,Microsoft.PowerShell.Commands.GetWinEventCommand

PS C:\> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4672; StartTime=(Get-Date).AddHours(-1)}
| ForEach-Object {$xml=[xml]$_.ToXml(); [PSCustomObject]@{TimeCreated=$_.TimeCreated; Account=$xml.Event
.EventData.Data[1].'#text'; LogonID=$xml.Event.EventData.Data[3].'#text'}} | Where-Object {$_.Account -n
otlike '*$'} | Format-Table -AutoSize

TimeCreated            Account         LogonID
-----------            -------         -------
03/11/2026 07:51:56 AM SYSTEM          0x3e7
03/11/2026 07:51:41 AM SYSTEM          0x3e7
03/11/2026 07:49:55 AM SYSTEM          0x3e7
03/11/2026 07:46:53 AM SYSTEM          0x3e7
03/11/2026 07:46:41 AM SYSTEM          0x3e7
03/11/2026 07:43:07 AM SYSTEM          0x3e7
03/11/2026 07:43:00 AM SYSTEM          0x3e7
03/11/2026 07:41:41 AM SYSTEM          0x3e7
03/11/2026 07:41:39 AM SYSTEM          0x3e7
03/11/2026 07:39:53 AM SYSTEM          0x3e7
03/11/2026 07:36:41 AM SYSTEM          0x3e7
03/11/2026 07:31:41 AM SYSTEM          0x3e7
03/11/2026 07:28:16 AM SYSTEM          0x3e7
03/11/2026 07:28:05 AM SYSTEM          0x3e7
03/11/2026 07:26:41 AM SYSTEM          0x3e7
03/11/2026 07:22:58 AM SYSTEM          0x3e7
03/11/2026 07:22:35 AM SYSTEM          0x3e7
03/11/2026 07:21:41 AM SYSTEM          0x3e7
03/11/2026 07:18:45 AM SYSTEM          0x3e7
03/11/2026 07:18:26 AM SYSTEM          0x3e7
03/11/2026 07:16:45 AM SYSTEM          0x3e7
03/11/2026 07:16:41 AM SYSTEM          0x3e7
03/11/2026 07:16:38 AM SYSTEM          0x3e7
03/11/2026 07:11:49 AM Anika Akter Nira 0x3209cfe
03/11/2026 07:11:38 AM SYSTEM          0x3e7
03/11/2026 07:11:21 AM SYSTEM          0x3e7
03/11/2026 06:56:17 AM SYSTEM          0x3e7
03/11/2026 06:56:12 AM SYSTEM          0x3e7
03/11/2026 06:56:12 AM SYSTEM          0x3e7
03/11/2026 06:56:12 AM SYSTEM          0x3e7
03/11/2026 06:55:45 AM SYSTEM          0x3e7
03/11/2026 06:55:22 AM SYSTEM          0x3e7
```

Explanation:

- ID=4672 identifies network share access events.
- Extracts account information.
- Identifies the source IP address.
- Displays the accessed network share name.

Exercise 14: Detect Special Privilege Assignments

Objective: To identify accounts receiving special system privileges.

Command:*Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4672; StartTime=(Get-Date).AddHours(-1)} | ForEach-Object { $xml=[xml]$_.ToXml(); [PSCustomObject]@{TimeCreated=$_.TimeCreated; Account=$xml.Event.EventData.Data[1].'#text'; LogonID=$xml.Event.EventData.Data[3].'#text'} } | Where-Object { $_.Account -notlike '*$' } | Format-Table –AutoSize*

```
PS C:\> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4672; StartTime=(Get-Date).AddHours(-1)}
| ForEach-Object {$xml=[xml]$_.ToXml(); [PSCustomObject]@{TimeCreated=$_.TimeCreated; Account=$xml.Event
.EventData.Data[1].'#text'; LogonID=$xml.Event.EventData.Data[3].'#text'}} | Where-Object {$_.Account -n
otlike '*$'} | Format-Table -AutoSize

TimeCreated            Account          LogonID
-----------            -------          -------
03/11/2026 07:51:56 AM SYSTEM           0x3e7
03/11/2026 07:51:41 AM SYSTEM           0x3e7
03/11/2026 07:49:55 AM SYSTEM           0x3e7
03/11/2026 07:46:53 AM SYSTEM           0x3e7
03/11/2026 07:46:41 AM SYSTEM           0x3e7
03/11/2026 07:43:07 AM SYSTEM           0x3e7
03/11/2026 07:43:00 AM SYSTEM           0x3e7
03/11/2026 07:41:41 AM SYSTEM           0x3e7
03/11/2026 07:41:39 AM SYSTEM           0x3e7
03/11/2026 07:39:53 AM SYSTEM           0x3e7
03/11/2026 07:36:41 AM SYSTEM           0x3e7
03/11/2026 07:31:41 AM SYSTEM           0x3e7
03/11/2026 07:28:16 AM SYSTEM           0x3e7
03/11/2026 07:28:05 AM SYSTEM           0x3e7
03/11/2026 07:26:41 AM SYSTEM           0x3e7
03/11/2026 07:22:58 AM SYSTEM           0x3e7
03/11/2026 07:22:35 AM SYSTEM           0x3e7
03/11/2026 07:21:41 AM SYSTEM           0x3e7
03/11/2026 07:18:45 AM SYSTEM           0x3e7
03/11/2026 07:18:26 AM SYSTEM           0x3e7
03/11/2026 07:16:45 AM SYSTEM           0x3e7
03/11/2026 07:16:41 AM SYSTEM           0x3e7
03/11/2026 07:16:38 AM SYSTEM           0x3e7
03/11/2026 07:11:49 AM Anika Akter Nira 0x3209cfe
03/11/2026 07:11:38 AM SYSTEM           0x3e7
03/11/2026 07:11:21 AM SYSTEM           0x3e7
03/11/2026 06:56:17 AM SYSTEM           0x3e7
03/11/2026 06:56:12 AM SYSTEM           0x3e7
03/11/2026 06:56:12 AM SYSTEM           0x3e7
03/11/2026 06:56:12 AM SYSTEM           0x3e7
03/11/2026 06:55:45 AM SYSTEM           0x3e7
03/11/2026 06:55:22 AM SYSTEM           0x3e7
```

Explanation:

- ID=4672 identifies special privilege assignment events.

- Retrieves events from the last hour.
- Extracts account information.
- Identifies the Logon ID.
- Filters out computer accounts ending with $.

Exercise 15: Create a Complete Security Analysis Script

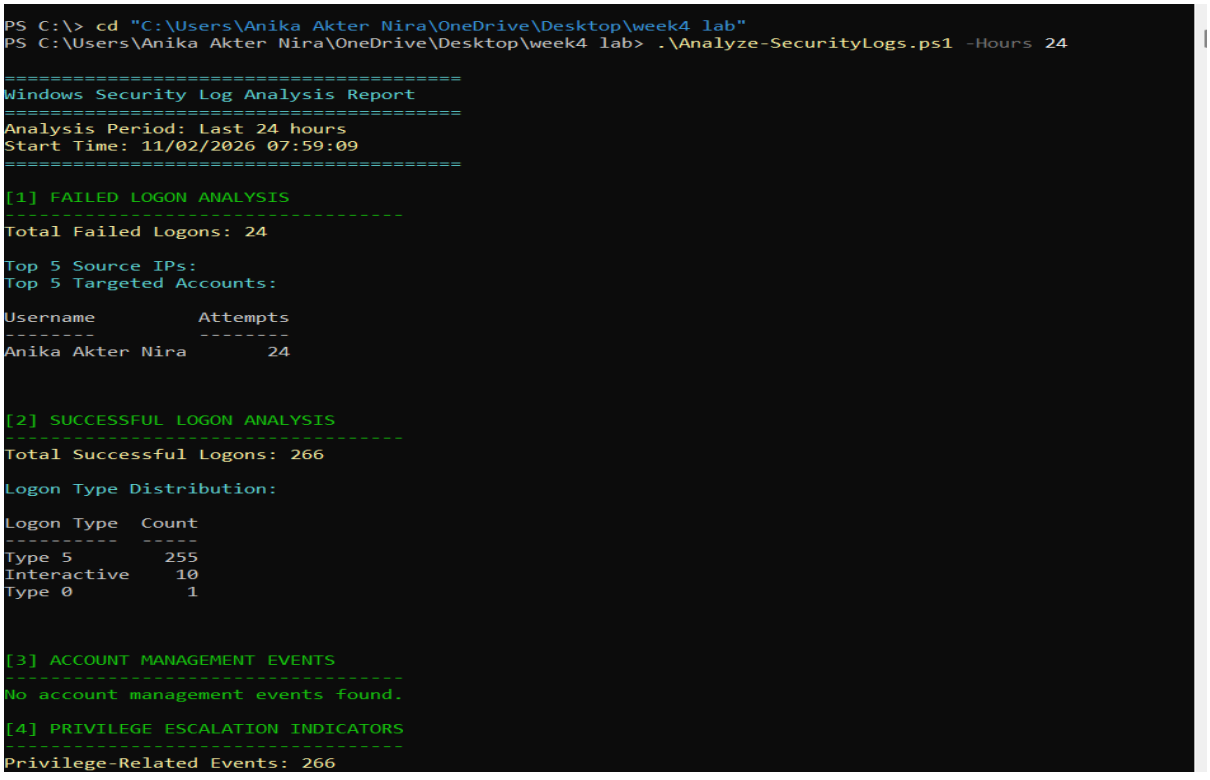
Objective: To create a PowerShell script that analyzes Windows Security Event Logs and identifies suspicious activities.

Command:

notepad Analyze-SecurityLogs.ps1
Copy the analysis script into the file and save it.

Run Command:

.\Analyze-SecurityLogs.ps1 -Hours 24



Explanation:

- Analyze-SecurityLogs.ps1 is the security analysis script.

- -Hours 24 specifies the analysis period.
- The script analyzes failed logons, successful logons, account management events, and privilege-related events.
- It displays a security analysis summary.

Exercise 16:Export to CSV

Objective: To export failed logon information into a CSV file for further analysis.

Command:

Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=(Get-Date).AddDays(-1)} | ForEach-Object { [PSCustomObject]@{TimeCreated=$_.TimeCreated; SourceIP=$_.Properties[19].Value; TargetUser=$_.Properties[5].Value; LogonType=$_.Properties[8].Value; FailureReason=$_.Properties[13].Value} } | Export-Csv -Path "FailedLogons.csv" -NoTypeInformation

```
========================================

PS C:\Users\Anika Akter Nira\OneDrive\Desktop\week4 lab> cd /
PS C:\> Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=(Get-Date).AddDays(-1)} |
 ForEach-Object {[PSCustomObject]@{TimeCreated=$_.TimeCreated; SourceIP=$_.Properties[19].Value; TargetU
ser=$_.Properties[5].Value; LogonType=$_.Properties[8].Value; FailureReason=$_.Properties[13].Value}} |
Export-Csv -Path "FailedLogons.csv" -NoTypeInformation; Write-Host "Exported to FailedLogons.csv" -Foreg
roundColor Green
Exported to FailedLogons.csv
```

Explanation:

- Retrieves failed logon events from the last 24 hours.
- Extracts time, source IP, username, logon type, and failure reason.
- Export-Csv saves the information into a CSV file.
- The file is named FailedLogons.csv.

Exercise 17:Export to HTML Report

Objective: To generate an HTML report containing failed logon information.

Command:

Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=(Get-Date).AddDays(-1)} | ForEach-Object

{ [PSCustomObject]@{TimeCreated=$_.TimeCreated; SourceIP=$_.Properties[19].Value; TargetUser=$_.Properties[5].Value; LogonType=$_.Properties[8].Value} }; $HTML=$FailedLogons | ConvertTo-Html -Title "Failed Logon Report" -PreContent "<h1>Failed Logon Analysis</h1> Generated: $(Get-Date) "; $HTML | Out-File "FailedLogons.html"

```
PS C:\> $FailedLogons = Get-WinEvent -FilterHashtable @{LogName='Security'; ID=4625; StartTime=(Get-Date
).AddDays(-1)} | ForEach-Object {[PSCustomObject]@{TimeCreated=$_.TimeCreated; SourceIP=$_.Properties[19
].Value; TargetUser=$_.Properties[5].Value; LogonType=$_.Properties[8].Value}}; $HTML = $FailedLogons |
ConvertTo-Html -Title "Failed Logon Report" -PreContent "<h1>Failed Logon Analysis</h1><p>Generated: $(G
et-Date)</p>"; $HTML | Out-File "FailedLogons.html"; Write-Host "Exported to FailedLogons.html" -Foregro
undColor Green
Exported to FailedLogons.html
PS C:\>
```

Explanation:

- Retrieves failed logon events.
- Extracts important login information.
- ConvertTo-Html converts the data into HTML format.
- Out-File saves the report as FailedLogons.html.
- The report can be opened in a web browser.

6. Security Event ID Summary

The following table summarizes the important event IDs used in this lab.

Event ID	Security Activity
4624	Successful login
4625	Failed login
4648	Explicit credential usage
4672	Special privilege assignment
4732	Local group membership change
5140	Network share access

7. Expected Outcomes

After completing this lab, students should be able to:

1. Access Windows Security Event Logs using PowerShell.
2. Identify successful and failed login attempts.

3. Analyze suspicious IP addresses and targeted accounts.
4. Understand different Windows logon types.
5. Investigate explicit credential usage.
6. Monitor network share access.
7. Detect possible privilege escalation activities.
8. Generate CSV and HTML security reports.

8. Conclusion

In this lab, we learned how to use Windows PowerShell for Windows Security Event Log analysis.

We investigated successful and failed login attempts, analyzed suspicious authentication activities, monitored network share access, and identified important security events.We also learned how to export security information into CSV and HTML reports.

Overall, this lab provided practical knowledge of Windows security monitoring and helped us understand how PowerShell can be used to investigate potential security threats.